Genetic and social risk factors for type 2 diabetes health disparities: a test of the Rose hypothesis

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OBJECTIVE

To evaluate the relative contributions of genetic and environmental risk factors to individual disease cases versus population disease disparites for type 2 diabetes (T2D) in the United Kingdom (UK)

ROSE HYPOTHESIS

In his seminal 1985 paper 'Sick individuals and sick populations', epidemiologist Geoffrey Arthur Rose CBE considered whether and how the causes of individual disease cases relate to the causes of population disease disparities [1]. He predicted that:

(1) Since genetic variation is greater within than between populations, genetic risk factors should be associated with individual cases but not population disparities

(2) Since environmental exposure variation is greater between than within populations, environmental risk factors should be associated with population disparities but not individual cases

STUDY COHORT AND T2D DISPARITIES

	Full cohort (n = 26,912) (100%)	Asian (n = 9,372) (34.82%)	Black (n = 7,540) (28.01%)	White (n = 10,000) (37.15%)
Mean age years (SD)	54.2 (8.42)	53.30 (8.45)	51.89 (8.05)	56.77 (7.99)
Female (%)	13,960 (51.87)	5,064 (54.10)	4,307 (57.12)	5,345 (53.45)
Male (%)	12,952 (48.12)	4,308 (45.90)	3,233 (42.80)	4,655 (46.55)
T2D Prev. (%)	15.52%	23.34%	16.64%	7.35%

Table 1 : Study Cohort derived from UK Biobank [2]. T2D prevelance is highest for Asian followed by Black and White ethinic groups

VARIANCE COMPONENTS

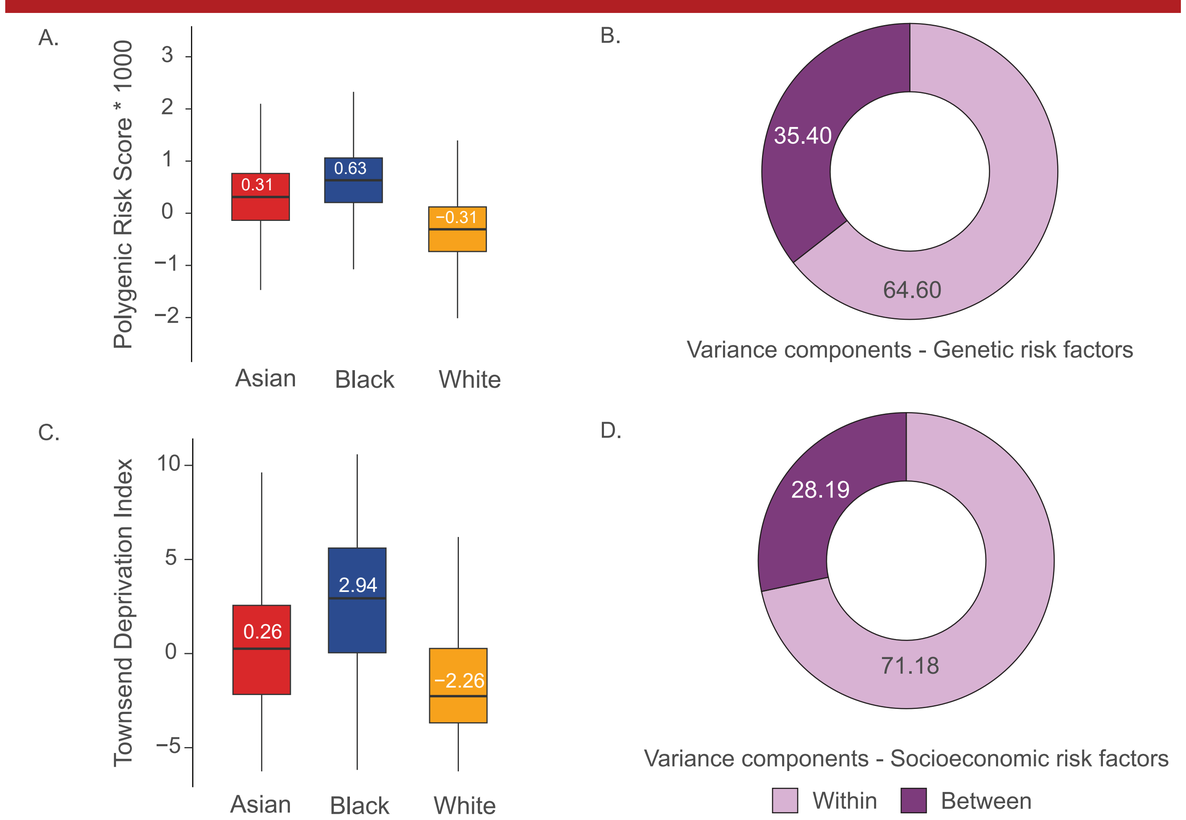

Figure 1 : (A) Box-plot distributions of polygenic risk scores (PRS) for Asian (red), Black (blue), and White (orange) ethnic groups. (B) PRS variance components within and between ethnic groups. (C) Box-plot distributions of Townsend index (TI) values for Asian (red), Black (blue), and White (orange) ethnic groups. (D) TI variance components within and between ethnic groups.

T2D INDIVIDUAL RISK

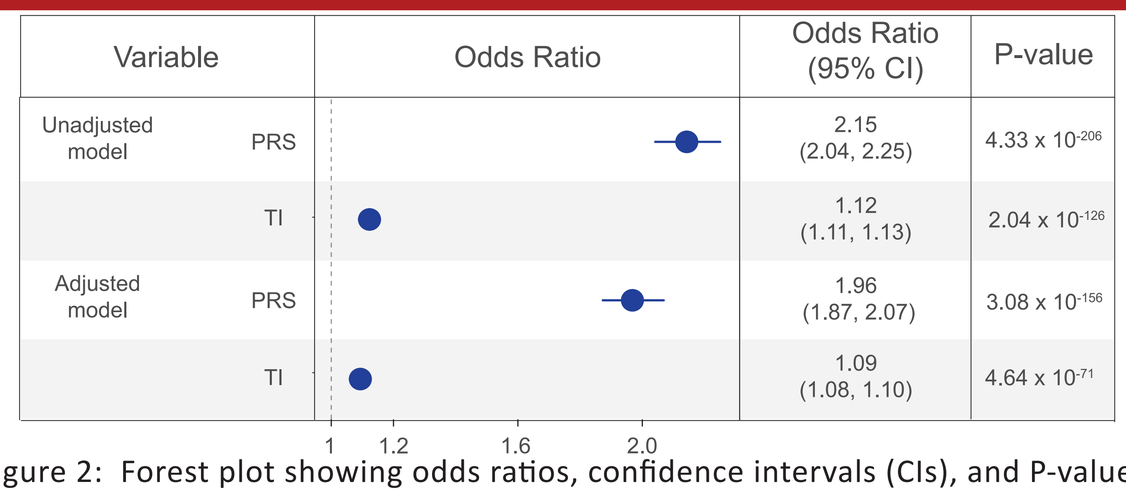

Figure 2: Forest plot showing odds ratios, confidence intervals (CIs), and P-values for PRS and TI associations with T2D. Model specifications (logistic regression): T2D ~ PRS + age + sex (unadjusted), T2D ~ TI + age + sex (unadjusted), T2D ~ TI + PRS + age + sex (adjusted). Both PRS and TI are positively associated with an individual's risk of T2D

T2D POPULATION DISPARITIES

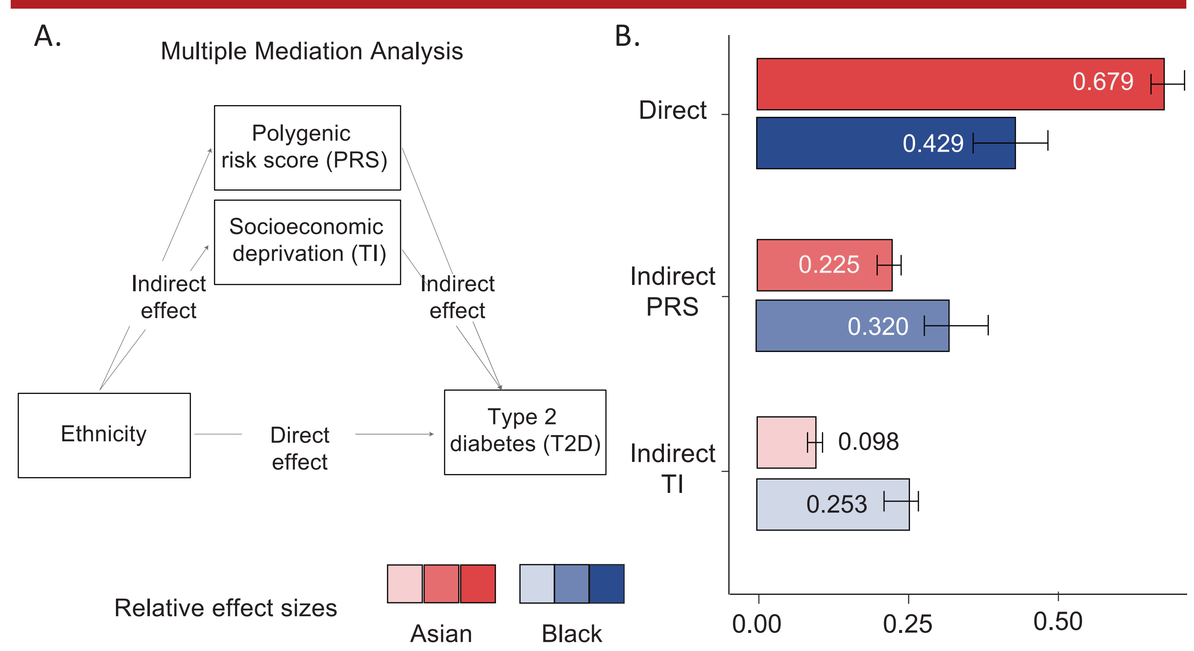

Figure 3: (A) Directed acyclic graph (DAG) indicating the outcome (T2D), predictor (ethnicity), and mediators (PRS and TI). Age and sex are included as covariates.[3] (B) Relative effect sizes are shown for the direct effect and the indirect effects mediated by PRS and TI, for Asian (red) and Black (blue) groups.

SUMMARY

(1) Contrary to Rose's predictions, both genetic and socioenvironmental risk factors show far greater variation within rather than between UK ethnic groups

(2) Both classes of risk factors are associated with individual-level T2D cases and mediate the observed T2D prevalence differences between ethnic groups

REFERENCES

1.Rose G. (2001). Sick individuals and sick populations. International journal of epidemiology, 30(3), 427–434. https://doi.org/10.1093/ije/30.3.427

2. Sudlow, C., Gallacher, J., Allen, N., Beral, V., Burton, P., Danesh, J., Downey, P., Elliott, P., Green, J., Landray, M., Liu, B., Matthews, P., Ong, G., Pell, J., Silman, A., Young, A., Sprosen, T., Peakman, T., & Collins, R. (2015). UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. PLoS medicine, 12(3), e1001779. https://doi.org/10.1371/journal.pmed.1001779

3. Naimi, A. I., Schnitzer, M. E., Moodie, E. E., & Bodnar, L. M. (2016). Mediation Analysis for Health Disparities Research. American journal of epidemiology, 184(4), 315–324. https://doi.org/10.1093/aje/kwv329